

Physical Chemistry (II)

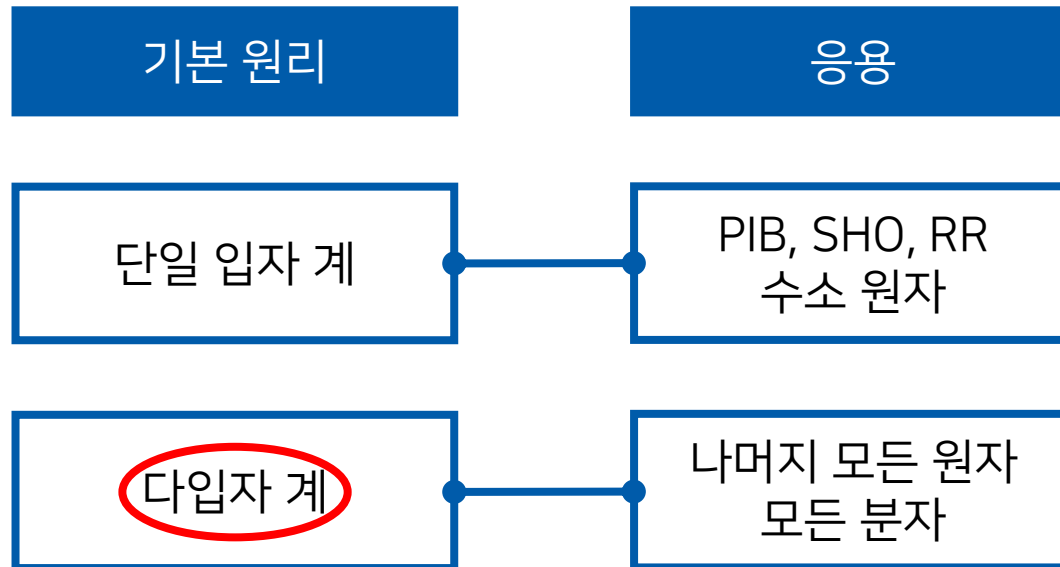
Lecture 15

Variational Method

Helium Atom



Where Are We?



Issues with Helium

- **Three-body problem:** not exactly solvable
 - Need approximation methods (e.g. perturbation theory)
- **Two electrons:** two identical particles
 - Meaning of wavefunctions
 - Indistinguishability: non-classical behaviors

Two-Particle Wavefunction

- $\Psi(\vec{r}_1, \vec{r}_2, t)$: a complex function of ...
 - particle 1 position $\vec{r}_1$
 - particle 2 position $\vec{r}_2$
 - time t
- $|\Psi(\vec{r}_1, \vec{r}_2, t)|^2$: probability of finding particle 1 at point $\vec{r}_1$ and particle 2 at point $\vec{r}_2$ at time t

Identical Particles

- Indistinguishability: one cannot tell where a specific particle is

$$|\psi(\vec{r}_a, \vec{r}_b)|^2 = |\psi(\vec{r}_b, \vec{r}_a)|^2$$

- Fermions vs. bosons
 - Fermions: half-integer spin (electron, proton, neutron, ...)

$$\psi(\vec{r}_a, \vec{r}_b) = -\psi(\vec{r}_b, \vec{r}_a) \text{ (antisymmetric)}$$

- Bosons: integer spin (photon, helium nucleus, ...)

$$\psi(\vec{r}_a, \vec{r}_b) = \psi(\vec{r}_b, \vec{r}_a) \text{ (symmetric)}$$

Spin Wavefunction

- For two spins, we can write its wavefunction as

$$|m_{s,1}\rangle_1 |m_{s,2}\rangle_2$$

- Indistinguishability issue
 - $|+1/2\rangle_1 |+1/2\rangle_2$: indistinguishable, symmetric
 - $|-1/2\rangle_1 |-1/2\rangle_2$: indistinguishable, symmetric
 - $|+1/2\rangle_1 |-1/2\rangle_2$ or $|-1/2\rangle_1 |+1/2\rangle_2$: distinguishable!

How can we make “indistinguishable” spin wavefunctions?

Last class

Addition of Spin Angular Momenta

- Consider two spins: $|\pm 1/2\rangle_1$ and $|\pm 1/2\rangle_2$
- Define the total spin angular momentum $\vec{S}_{\text{tot}} = \vec{S}_1 + \vec{S}_2$:

$$\hat{S}_{\text{tot}}^2 |s_{\text{tot}} m_{s,\text{tot}}\rangle = s_{\text{tot}}(s_{\text{tot}} + 1)\hbar^2 |s_{\text{tot}} m_{s,\text{tot}}\rangle$$

$$\hat{S}_{z,\text{tot}} |s_{\text{tot}} m_{s,\text{tot}}\rangle = m_{s,\text{tot}}\hbar |s_{\text{tot}} m_{s,\text{tot}}\rangle$$

- Total spin quantum numbers

$$s_{\text{tot}} = (s_1 + s_2), (s_1 + s_2 - 1), \dots, |s_1 - s_2|$$

$$m_{s,\text{tot}} = -s_{\text{tot}}, -(s_{\text{tot}} - 1), \dots, s_{\text{tot}}$$

Spin Wavefunctions

- Triplet: symmetric

- $|s_{\text{tot}} = 1, m_{s,\text{tot}} = 1\rangle_{\text{tot}} = |+\frac{1}{2}\rangle_1 |+\frac{1}{2}\rangle_2$

- $|1, -1\rangle_{\text{tot}} = |-\frac{1}{2}\rangle_1 |-\frac{1}{2}\rangle_2$

- $|1, 0\rangle_{\text{tot}} = \frac{1}{\sqrt{2}} (|+\frac{1}{2}\rangle_1 |-\frac{1}{2}\rangle_2 + |-\frac{1}{2}\rangle_1 |+\frac{1}{2}\rangle_2)$

- Singlet: antisymmetric

- $|0, 0\rangle_{\text{tot}} = \frac{1}{\sqrt{2}} (|+\frac{1}{2}\rangle_1 |-\frac{1}{2}\rangle_2 - |-\frac{1}{2}\rangle_1 |+\frac{1}{2}\rangle_2)$

Last class

Total Two-Electron Wavefunctions

$$\psi(\vec{r}_1, \vec{r}_2) |s_{\text{tot}} m_{s,\text{tot}}\rangle_{\text{tot}}$$

- The total wavefunction must be antisymmetric
- If the spin wavefunction is triplet (symmetric),
the spatial wavefunction ψ must be antisymmetric
- If the spin wavefunction is singlet (antisymmetric),
the spatial wavefunction ψ must be symmetric

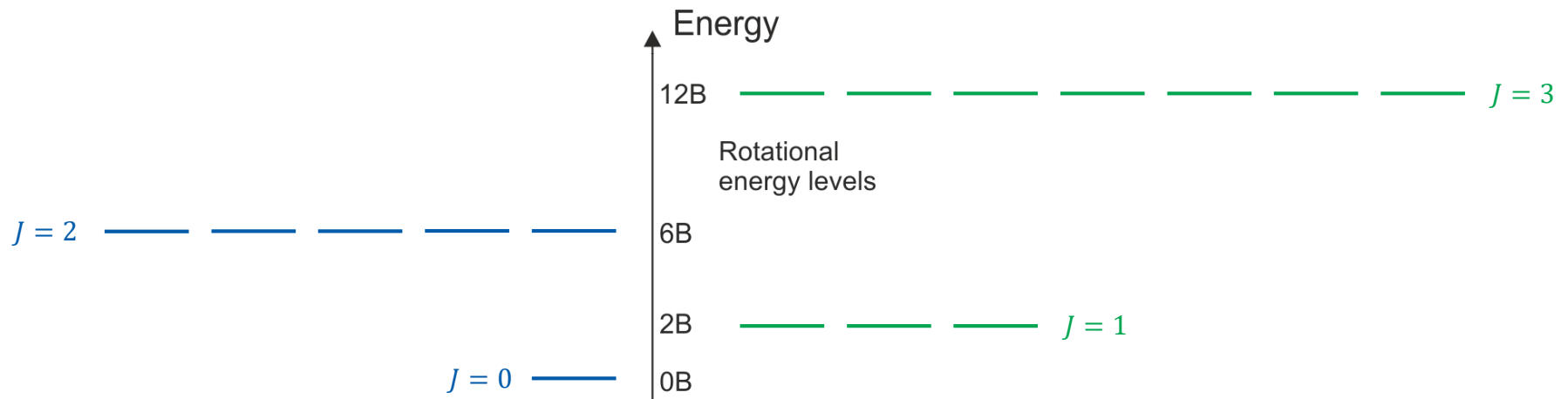
H₂ Molecule

- Consider only rotation and spin ($s = 1/2$ for a proton)
- Total wavefunction: $Y_J^m(\theta, \phi) |s_{\text{tot}} m_{s,\text{tot}}\rangle_{\text{tot}}$
 - Must be **antisymmetric**
- **Triplet** $|1 m_{s,\text{tot}}\rangle_{\text{tot}}$: symmetric (“ortho-hydrogen”)
 - Y_J^m must be antisymmetric: **odd J**
- **Singlet** $|0 0\rangle_{\text{tot}}$: antisymmetric (“para-hydrogen”)
 - Y_J^m must be symmetric: **even J**

Last class

Energy Levels

(Number of states) $\times$ 3
due to spin states



para-hydrogen
(singlet, $s_{\text{tot}} = 0$)



ortho-hydrogen
(triplet, $s_{\text{tot}} = 1$)

rare transition
(selection rule)

Approximation Methods

$\hat{H}\psi = E\psi$: No more exactly solvable

→ need approximation!

1. Perturbation theory
2. Variational Method

Variational Principle

Ground-state wavefunction ψ_0 and its energy E_0 satisfy

$$\hat{H}\psi_0 = E_0\psi_0$$

Multiplication by $\psi_0^*/\int \psi_0^*\psi_0 d\tau$ and integration lead to

$$E_0 = \frac{\int \psi_0^* \hat{H} \psi_0 d\tau}{\int \psi_0^* \psi_0 d\tau} = \frac{\langle \psi_0 | \hat{H} | \psi_0 \rangle}{\langle \psi_0 | \psi_0 \rangle}$$

For any function ϕ (“trial function”), the following holds:

$$E_\phi = \frac{\int \phi^* \hat{H} \phi d\tau}{\int \phi^* \phi d\tau} = \frac{\langle \phi | \hat{H} | \phi \rangle}{\langle \phi | \phi \rangle} \geq E_0$$

Proof

Since the set of wavefunctions $\{\psi_i\}$ is complete,

$$\phi = \sum_i c_i \psi_i = c_1 \psi_1 + c_2 \psi_2 + \dots$$

$$\int \phi^* \phi \, d\tau = \int (c_1^* \psi_1^* + c_2^* \psi_2^* + \dots)(c_1 \psi_1 + c_2 \psi_2 + \dots) \, d\tau$$

$$= \int (c_1^* c_1 \psi_1^* \psi_1 + c_2^* c_2 \psi_2^* \psi_2 + \dots + c_1^* c_2 \psi_1^* \psi_2 + \dots) \, d\tau$$

$$= c_1^* c_1 \int \cancel{\psi_1^* \psi_1} \, d\tau + c_2^* c_2 \int \cancel{\psi_2^* \psi_2} \, d\tau + \dots + c_1^* c_2 \int \cancel{\psi_1^* \psi_2} \, d\tau + \dots$$

1
1
0

Proof

$$\phi = \sum_i c_i \psi_i = c_1 \psi_1 + c_2 \psi_2 + \dots$$

$$\int \phi^* \phi \, d\tau = c_1^* c_1 + c_2^* c_2 + \dots = \sum_i |c_i|^2$$

$$\int \phi^* \hat{H} \phi \, d\tau = \int (c_1^* \psi_1^* + c_2^* \psi_2^* + \dots) \hat{H} (c_1 \psi_1 + c_2 \psi_2 + \dots) \, d\tau$$

$$= \int (c_1^* \psi_1^* + c_2^* \psi_2^* + \dots) (c_1 E_1 \psi_1 + c_2 E_2 \psi_2 + \dots) \, d\tau$$

$$= c_1^* c_1 E_1 + c_2^* c_2 E_2 + \dots = \sum_i |c_i|^2 E_i$$

Proof

$$\phi = \sum_i c_i \psi_i = c_1 \psi_1 + c_2 \psi_2 + \dots$$

$$\int \phi^* \phi \, d\tau = \sum_i |c_i|^2, \quad \int \phi^* \hat{H} \phi \, d\tau = \sum_i |c_i|^2 E_i$$

Hence,

$$E_\phi = \frac{\int \phi^* \hat{H} \phi \, d\tau}{\int \phi^* \phi \, d\tau} = \frac{\sum_i |c_i|^2 E_i}{\sum_i |c_i|^2} \geq \frac{\sum_i |c_i|^2 E_0}{\sum_i |c_i|^2} = E_0$$

Trial Function

$$E_{\phi} \geq E_0$$

- May contain parameters (variational parameters)

$$\phi = f(\alpha, \beta, \gamma, \dots)$$

Minimize $E_{\phi}(\alpha, \beta, \gamma, \dots)$!

- May be a linear combination of basis functions

$$\phi = c_1 f_1 + c_2 f_2 + \dots + c_N f_N$$

Minimize $E_{\phi}(c_1, c_2, \dots, c_N)$!

Example: Hydrogen Atom

$$\hat{H} = -\frac{\hbar^2}{2\mu} \nabla^2 - \frac{e^2}{4\pi\epsilon_0 r}$$

- Trial function: $\phi(\alpha) = e^{-\alpha r}$ (α : variational parameter)

$$\int \phi^* \hat{H} \phi \, d\tau = \frac{3\hbar^2 \pi^{3/2}}{4\sqrt{2}\mu\alpha^{1/2}} - \frac{e^2}{4\epsilon_0 \alpha}$$

$$\int \phi^* \phi \, d\tau = \left(\frac{\pi}{2\alpha}\right)^{3/2}$$

Example: Hydrogen Atom

$$E_{\phi}(\alpha) = \frac{\int \phi^* \hat{H} \phi \, d\tau}{\int \phi^* \phi \, d\tau} = \frac{3\hbar^2 \alpha}{2\mu} - \frac{e^2}{\epsilon_0} \left(\frac{\alpha}{2\pi^3} \right)^{1/2}$$

Minimizing $E_{\phi}(\alpha)$ by differentiation, we obtain

$$E_{\phi,\min} = -\frac{4}{3\pi} \left(\frac{\mu e^4}{16\pi^2 \epsilon_0^2 \hbar^2} \right) = -0.424 \left(\frac{\mu e^4}{16\pi^2 \epsilon_0^2 \hbar^2} \right)$$

- True value:

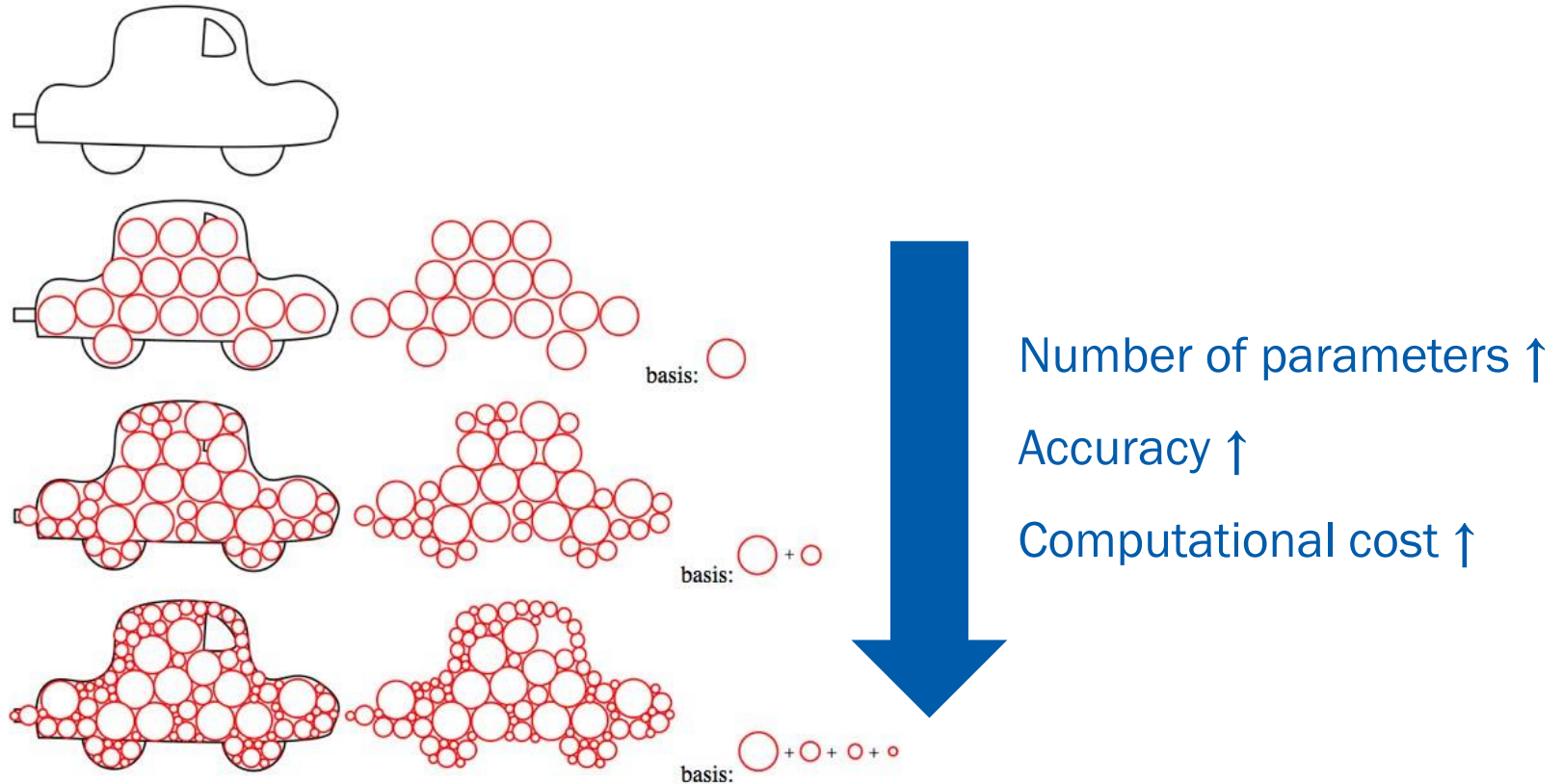
$$E_0 = -\frac{1}{2} \left(\frac{\mu e^4}{16\pi^2 \epsilon_0^2 \hbar^2} \right) = -0.500 \left(\frac{\mu e^4}{16\pi^2 \epsilon_0^2 \hbar^2} \right)$$

H-Atom Ground-State Energy

$$\phi(\{c_j, \alpha_j\}) = \sum_{j=1}^N c_j e^{-\alpha_j r^2}$$

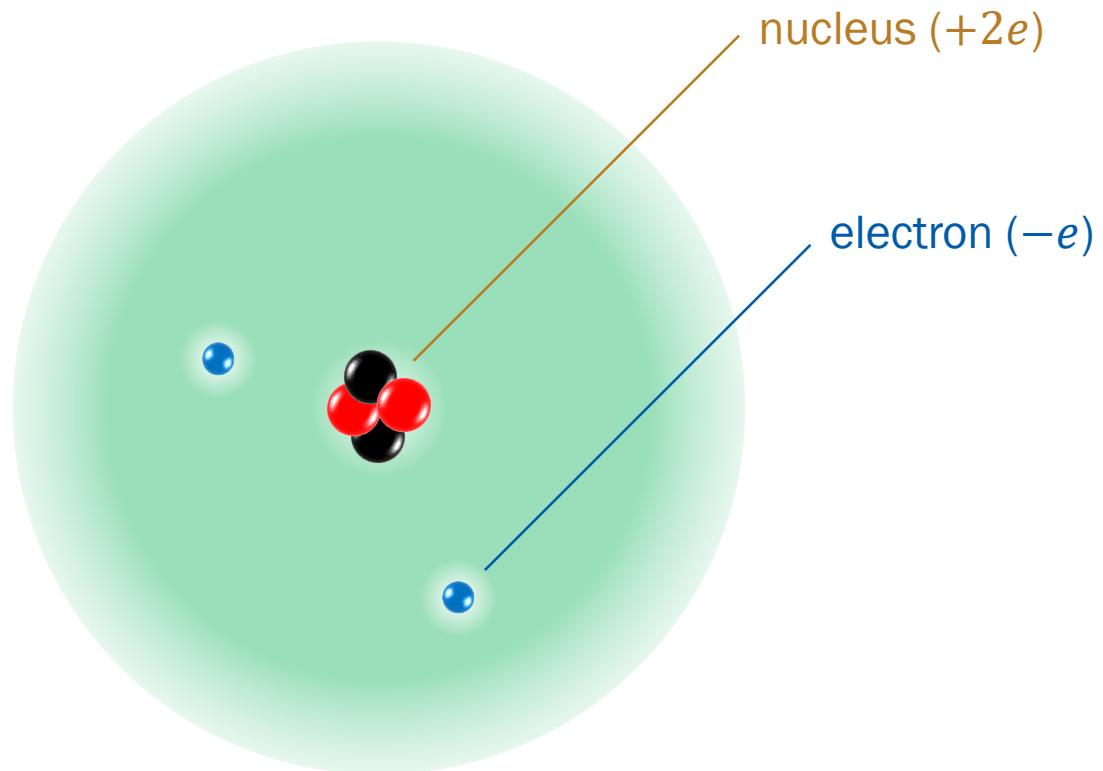
N	$E_{\phi, \min} / (m_e e^4 / 16 \pi^2 \varepsilon_0^2 \hbar^2)$
1	-0.42441
2	-0.48581
3	-0.49697
4	-0.49928
5	-0.49976
6	-0.49988
8	-0.49992
16	-0.49998

Number of Parameters



RS 7.1
M 7.9

Helium Atom



Hamiltonian of Helium Atom

$$\hat{H} = -\frac{\hbar^2}{2M} \nabla_{\text{nuc}}^2 - \frac{\hbar^2}{2m_e} \nabla_{e1}^2 - \frac{\hbar^2}{2m_e} \nabla_{e2}^2 + \frac{1}{4\pi\epsilon_0} \left(-\frac{2e^2}{|\vec{R} - \vec{r}_1|} - \frac{2e^2}{|\vec{R} - \vec{r}_2|} + \frac{e^2}{|\vec{r}_1 - \vec{r}_2|} \right)$$

Assuming that the nucleus is fixed at the origin ($\vec{R} = 0$),

$$\hat{H} = -\frac{\hbar^2}{2m_e} \nabla_{e1}^2 - \frac{\hbar^2}{2m_e} \nabla_{e2}^2 + \frac{1}{4\pi\epsilon_0} \left(-\frac{2e^2}{r_1} - \frac{2e^2}{r_2} + \frac{e^2}{|\vec{r}_1 - \vec{r}_2|} \right)$$

Hamiltonian of Helium Atom

$$\hat{H} = -\frac{\hbar^2}{2m_e}\nabla_{e1}^2 - \frac{\hbar^2}{2m_e}\nabla_{e2}^2 + \frac{1}{4\pi\epsilon_0}\left(-\frac{2e^2}{r_1} - \frac{2e^2}{r_2} + \frac{e^2}{|\vec{r}_1 - \vec{r}_2|}\right)$$

$$\hat{H} = \underbrace{-\frac{\hbar^2}{2m_e}\nabla_{e1}^2 - \frac{2e^2}{4\pi\epsilon_0 r_1}}_{\hat{H}_{H1}} - \underbrace{\frac{\hbar^2}{2m_e}\nabla_{e2}^2 - \frac{2e^2}{4\pi\epsilon_0 r_2}}_{\hat{H}_{H2}} + \underbrace{\frac{e^2}{4\pi\epsilon_0 |\vec{r}_1 - \vec{r}_2|}}_{\text{interelectronic repulsion}}$$

Let us start from “independent electrons”
(neglect the interelectronic repulsion)

Perturbation Theory (Ground State)

$$\hat{H}_0 = \hat{H}_{H1} + \hat{H}_{H2}, \quad \hat{H}' = \frac{e^2}{4\pi\epsilon_0|\vec{r}_1 - \vec{r}_2|}$$

$$\psi^{(0)}(\vec{r}_1, \vec{r}_2) = \psi_{1s}(\vec{r}_1)\psi_{1s}(\vec{r}_2)$$

$$E^{(0)} = \left(-\frac{1}{2}Z^2 - \frac{1}{2}Z^2 \right) \frac{m_e e^4}{16\pi^2 \epsilon_0^2 \hbar^2}$$

The first-order correction is

$$E^{(1)} = \int \psi^{(0)*} \hat{H}' \psi^{(0)} d\vec{r}_1 d\vec{r}_2 = \frac{5Z}{8} \frac{m_e e^4}{16\pi^2 \epsilon_0^2 \hbar^2}$$

Variational Method

Set the trial function as

$$\phi_0(\vec{r}_1, \vec{r}_2) = \psi_{1s}(\vec{r}_1; Z)\psi_{1s}(\vec{r}_2; Z) = \left(\frac{Z^3}{\pi a_0^3}\right) e^{-Zr_1/a_0} e^{-Zr_2/a_0}$$

The energy is given by

$$E(Z) = \frac{\int \phi_0^* \hat{H} \phi_0 d\vec{r}_1 d\vec{r}_2}{\int \phi_0^* \phi_0 d\vec{r}_1 d\vec{r}_2} = \frac{m_e e^4}{16\pi^2 \epsilon_0^2 \hbar^2} \left(Z^2 - \frac{27}{8} Z \right)$$

Minimization gives $Z_{\min} = 27/16$ and

$$E_{\min} = - \left(\frac{27}{16} \right)^2 \frac{m_e e^4}{16\pi^2 \epsilon_0^2 \hbar^2}$$

Hartree: Atomic Unit of Energy

$$E_h = \frac{m_e e^4}{16\pi^2 \epsilon_0^2 \hbar^2}$$

- Perturbation theory (first-order, ground state)

$$E = E^{(0)} + E^{(1)} = \left(-Z^2 + \frac{5}{8}Z \right) E_h = -2.75 E_h$$

- Variational method (1s orbitals)

$$E = -\left(\frac{27}{16} \right)^2 E_h = -2.8477 E_h$$

Better Trial Functions

$$\phi(\vec{r}_1, \vec{r}_2) = \frac{(Z')^{3/2} (Z'')^{3/2}}{\pi} (e^{-Z' r_1} e^{-Z'' r_2} + e^{-Z' r_2} e^{-Z'' r_1})$$

$$\rightarrow Z' = 1.19, Z'' = 2.18, E_\phi = -2.8757 E_h$$

$$\phi(\vec{r}_1, \vec{r}_2) = \frac{1 + b r_{12}}{N} e^{-Z'(r_1 + r_2)}$$

$$\rightarrow Z' = 1.849, b = 0.364, E_\phi = -2.892 E_h$$

Ground-State Energy of He

Method	Energy (E_h)	Method	Energy (E_h)
<i>Complete neglect of the interelectronic repulsion term</i>			-4.0000
<i>Perturbation calculations</i>		<i>Variational calculations*</i>	
First-order	-2.7500	1s orbitals (1)	-2.8477
Second-order	-2.9077	Eckart (2)	-2.8757
13 th -order	-2.90372433	Hartree-Fock (2)	-2.86168
		Hylleraas (10)	-2.90363
		Pekeris (1078)	-2.903724375
<i>Experimental value</i>			-2.9033

* the number in parentheses indicates the number of parameters

Total Wavefunction

The He ground-state spatial wavefunction is **symmetric**

$$\phi_0(\vec{r}_1, \vec{r}_2) = \psi_{1s}(\vec{r}_1; Z)\psi_{1s}(\vec{r}_2; Z)$$

$$\phi(\vec{r}_1, \vec{r}_2) = \frac{(Z')^{3/2}(Z'')^{3/2}}{\pi} (e^{-Z'r_1}e^{-Z''r_2} + e^{-Z'r_2}e^{-Z''r_1})$$

$$\phi(\vec{r}_1, \vec{r}_2) = \frac{1 + br_{12}}{N} e^{-Z'(r_1+r_2)}$$

The spin wavefunction must be **antisymmetric** (singlet!)

$$|0, 0\rangle_{\text{tot}} = \frac{1}{\sqrt{2}} (|+1/2\rangle_1|-1/2\rangle_2 - |-1/2\rangle_1|+1/2\rangle_2)$$

Excited State of Helium

- The first excited state of He: $(1s)^1(2s)^1$
- Simple approximation: each electron's spatial part is represented by an one-electron orbital $\phi_{1s}(\vec{r})$, $\phi_{2s}(\vec{r})$

We can construct

$$\Phi_1(\vec{r}_1, \vec{r}_2) = \phi_{1s}(\vec{r}_1)\phi_{2s}(\vec{r}_2), \Phi_2(\vec{r}_1, \vec{r}_2) = \phi_{2s}(\vec{r}_1)\phi_{1s}(\vec{r}_2)$$

but they are **NOT symmetric nor antisymmetric**

Symmetrize them!

Spatial Wavefunction

$$\Phi_1(\vec{r}_1, \vec{r}_2) = \phi_{1s}(\vec{r}_1)\phi_{2s}(\vec{r}_2), \Phi_2(\vec{r}_1, \vec{r}_2) = \phi_{2s}(\vec{r}_1)\phi_{1s}(\vec{r}_2)$$

We can add and subtract them to obtain

$$\Phi_+ = \frac{1}{\sqrt{2}} [\phi_{1s}(\vec{r}_1)\phi_{2s}(\vec{r}_2) + \phi_{2s}(\vec{r}_1)\phi_{1s}(\vec{r}_2)] \quad \text{symmetric}$$

$$\Phi_- = \frac{1}{\sqrt{2}} [\phi_{1s}(\vec{r}_1)\phi_{2s}(\vec{r}_2) - \phi_{2s}(\vec{r}_1)\phi_{1s}(\vec{r}_2)] \quad \text{antisymmetric}$$

Symmetric spatial wf $\rightarrow$ antisymmetric spin wf (singlet)

Antisymmetric spatial wf $\rightarrow$ symmetric spin wf (triplet)

Total Wavefunctions

- Triplet states

$$\Phi_{1,1} = \frac{1}{\sqrt{2}} [\phi_{1s}(\vec{r}_1)\phi_{2s}(\vec{r}_2) - \phi_{2s}(\vec{r}_1)\phi_{1s}(\vec{r}_2)]|1, 1\rangle_{\text{tot}}$$

$$\Phi_{1,0} = \frac{1}{\sqrt{2}} [\phi_{1s}(\vec{r}_1)\phi_{2s}(\vec{r}_2) - \phi_{2s}(\vec{r}_1)\phi_{1s}(\vec{r}_2)]|1, 0\rangle_{\text{tot}}$$

$$\Phi_{1,-1} = \frac{1}{\sqrt{2}} [\phi_{1s}(\vec{r}_1)\phi_{2s}(\vec{r}_2) - \phi_{2s}(\vec{r}_1)\phi_{1s}(\vec{r}_2)]|1, -1\rangle_{\text{tot}}$$

- Singlet state

$$\Phi_{0,0} = \frac{1}{\sqrt{2}} [\phi_{1s}(\vec{r}_1)\phi_{2s}(\vec{r}_2) + \phi_{2s}(\vec{r}_1)\phi_{1s}(\vec{r}_2)]|0, 0\rangle_{\text{tot}}$$

Energies of He(1s2s) States

$$\hat{H}_0 = \hat{H}_{H1} + \hat{H}_{H2}, \quad \hat{H}' = \frac{e^2}{4\pi\epsilon_0|\vec{r}_1 - \vec{r}_2|}$$

- The Hamiltonian does not apply on spin wavefunctions
- Zeroth-order energy

$$\begin{aligned} & \hat{H}_0[\phi_{1s}(\vec{r}_1)\phi_{2s}(\vec{r}_2) \pm \phi_{2s}(\vec{r}_1)\phi_{1s}(\vec{r}_2)]/\sqrt{2} \\ &= (E_{1s} + E_{2s})[\phi_{1s}(\vec{r}_1)\phi_{2s}(\vec{r}_2) \pm \phi_{2s}(\vec{r}_1)\phi_{1s}(\vec{r}_2)]/\sqrt{2} \end{aligned}$$

Energies of He(1s2s) States

$$\hat{H}_0 = \hat{H}_{H1} + \hat{H}_{H2}, \quad \hat{H}' = \frac{e^2}{4\pi\epsilon_0|\vec{r}_1 - \vec{r}_2|}$$

- First-order energy

$$E^{(1)} = \int \Phi_{\pm}^*(\vec{r}_1, \vec{r}_2) \hat{H}' \Phi_{\pm}(\vec{r}_1, \vec{r}_2) d\tau$$

$$E^{(1)} = \int \phi_{1s}^*(\vec{r}_1) \phi_{2s}^*(\vec{r}_2) \hat{H}' \phi_{1s}(\vec{r}_1) \phi_{2s}(\vec{r}_2) d\tau$$

$$\pm \int \phi_{1s}^*(\vec{r}_1) \phi_{2s}^*(\vec{r}_2) \hat{H}' \phi_{2s}(\vec{r}_1) \phi_{1s}(\vec{r}_2) d\tau$$

Coulomb and Exchange Integrals

$$E^{(1)} = \underbrace{\int \phi_{1s}^*(\vec{r}_1) \phi_{2s}^*(\vec{r}_2) \hat{H}' \phi_{1s}(\vec{r}_1) \phi_{2s}(\vec{r}_2) d\tau}_{\text{Coulomb integral } J_{1s2s}} \pm \underbrace{\int \phi_{1s}^*(\vec{r}_1) \phi_{2s}^*(\vec{r}_2) \hat{H}' \phi_{2s}(\vec{r}_1) \phi_{1s}(\vec{r}_2) d\tau}_{\text{Exchange integral } K_{1s2s}}$$

- **Coulomb integral:** repulsion between 1s and 2s electrons
- **Exchange integral:** repulsion between overlap density; due to the antisymmetric nature of electrons

Energies of He(1s2s) States

$$E = E^{(0)} + E^{(1)} = E_{1s} + E_{2s} + J_{1s2s} \pm K_{1s2s}$$

- Triplet state: $\Phi_- |1 m_s\rangle$

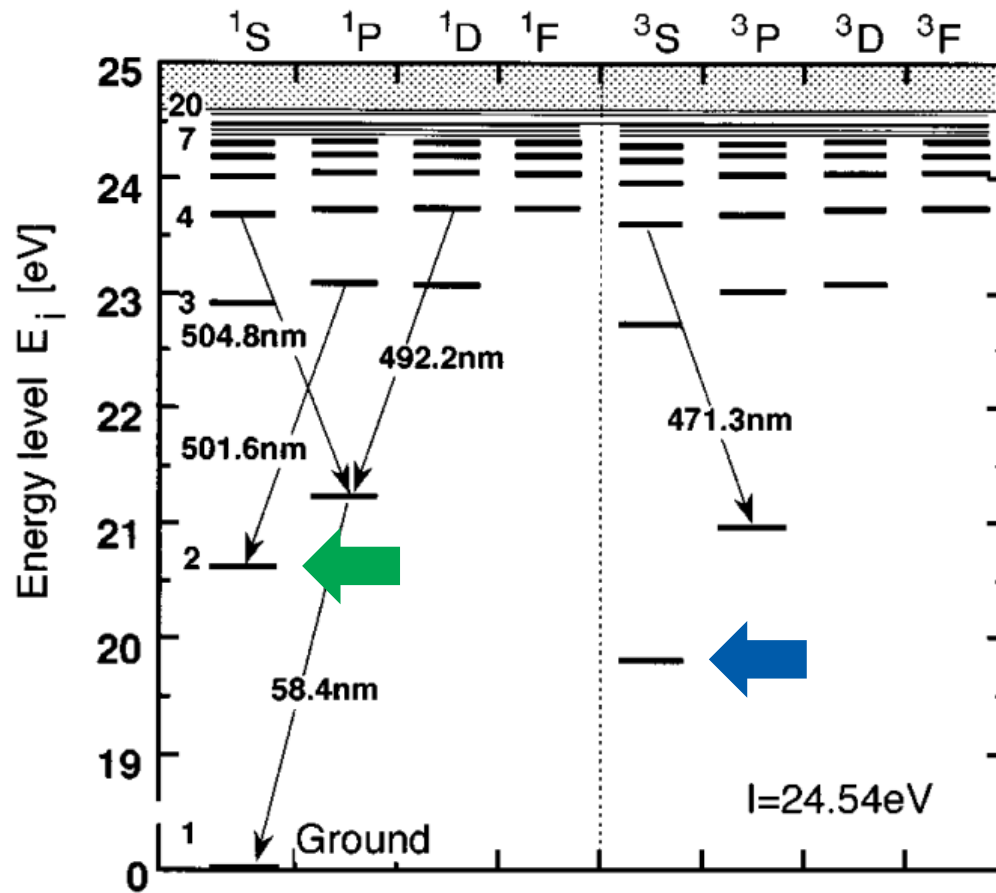
$$E = E_{1s} + E_{2s} + J_{1s2s} - K_{1s2s}$$

- Singlet state: $\Phi_+ |0 0\rangle$

$$E = E_{1s} + E_{2s} + J_{1s2s} + K_{1s2s}$$

The triplet states are more stable than the singlet state

Energy Levels of Helium



(Image: Sasaki et al., *Rev. Sci. Instrum.* 1996)